Disaster Relief Information Extraction Using Natural Language Processing and Machine Learning

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***Abstract*—Timely and accurate information is crucial for disaster relief. Social media, particularly Twitter, provides real-time updates, but filtering relevant (on-topic) disaster-related information is challenging. This study applies Natural Language Processing (NLP) techniques to classify tweets using the CrisisLexT6 dataset. We employed Bag of Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF) for feature extraction and evaluated six classifiers. For BoW, our results show that Logistic Regression (94.82%) achieved the highest accuracy, followed by Random Forest (94.52%) and SVM (94.12%). Naïve Bayes performed well (91.50%) but was outperformed by other models, while KNN had the lowest accuracy (84.17%), likely due to its inefficiency with high-dimensional text data. For TF-IDF, our results show that SVM achieved the highest accuracy (94.65%), followed by Random Forest (94.59%) and Logistic Regression (93.31%). Naïve Bayes performed well (81.87%) but was outperformed by other models, while KNN had the lowest accuracy (63.27%). This study highlights that machine learning-based NLP models can enhance disaster tweet classification, improving crisis response. Future work will explore deep learning models like LSTMs and Transformers for better accuracy.**

***Index Terms*—Machine Learning, NLP, Text Classification, Disaster Relief, Logistic Regression, SVM, Random Forest, Naïve Bayes, KNN, CrisisLexT6**

I. Introduction

Social media platforms have become an important medium for sharing information during disaster events. Users frequently post updates related to damaged infrastructure, affected populations, emergency needs, and ongoing relief efforts. Among various social networking platforms, Twitter is particularly prominent due to its real-time communication capabilities and wide public reach. During emergencies, this rapid information flow can be valuable; however, the massive number of tweets generated within a short time frame makes manual analysis impractical. As a result, relevant disaster-related information is often buried under irrelevant or misleading content.

To address this challenge, automated approaches based on Natural Language Processing (NLP) and Machine Learning (ML) have gained significant attention. These techniques enable efficient filtering and classification of tweets, allowing emergency responders, government agencies, and humanitarian organizations to identify critical information more quickly. Automated tweet classification can assist in prioritizing response actions and improving situational awareness during disaster scenarios.

In this study, the CrisisLexT6 dataset is used to train and evaluate multiple machine learning models for disaster tweet classification. Textual data is transformed into numerical representations using two feature extraction methods: Bag of Words (BoW) and Term Frequency–Inverse Document Frequency (TF-IDF). Six machine learning classifiers are then applied to the extracted features, and their performance is evaluated using accuracy and additional evaluation metrics.

The objectives of this research are threefold:

- To develop an automated system for classifying disaster-related tweets.
- To examine the impact of different feature extraction techniques, namely BoW and TF-IDF, on classification performance.
- To analyze and compare the effectiveness of various machine learning classifiers for disaster tweet classification.

The experimental results indicate that Support Vector Machine (SVM), Random Forest, and Logistic Regression classifiers achieve comparatively better performance than other models considered in this study. These findings demonstrate the potential of machine learning-based approaches in supporting disaster response and decision-making processes. Future work will focus on exploring deep learning techniques, such as Long Short-Term Memory (LSTM) networks and Transformer-based models including BERT, to further enhance classification accuracy and robustness.

II. Literature Review

1) Olteanu et al. propose a crisis lexicon to improve the recall of crisis-related tweets. Experiments show that adding the lexicon to manually chosen keywords leads to substantial improvements in recall while preserving the original distribution of message types.
2) This paper reviews methods for processing social media data in emergencies, focusing on classification, summarization, and decision-making. It highlights existing approaches, their effectiveness, and future directions for crisis response.

3) The paper proposes a transformer-based encoder model to classify crisis-related tweets into categories like "relevant" and "irrelevant." It outperforms baseline models like CNN, BiLSTM, and BERT on datasets such as CrisisNLP and CrisisLexT6, achieving an accuracy of 92.87% and an F1-score of 90.61%.
4) The paper reviews ML and deep learning methods (e.g., SVM, LSTM) for identifying actionable tweets during emergencies, with reported accuracies ranging from 70% to 89%. It outlines challenges like data scarcity and ambiguity in defining actionability, and suggests future directions like multimodal inputs and domain adaptation.

III. Methodology

A. Dataset

We utilize the CrisisLexT6 dataset, which contains 60,000 tweets from different disaster events. Each tweet is labeled as either on-topic (relevant to disasters, 32,000 tweets) or off-topic (irrelevant to disasters, 28,000 tweets), as shown in Table I.

TABLE I
Dataset Distribution

Label	Number of Tweets
On-Topic	32,462
Off-Topic	27,420

We preprocessed the dataset and split it into training and testing sets, as shown in Table II.

TABLE II
Dataset Splitting

Set	Label	Number of Tweets
Training	On-Topic	25,969
	Off-Topic	22,096
Testing	On-Topic	6,493
	Off-Topic	5,524

B. Proposed System

Our proposed system focuses on classifying tweets into disaster-relevant and irrelevant categories. The system follows a structured pipeline as illustrated in Fig. 1.

C. Data Preprocessing

Before training machine learning models, preprocessing is required to clean and standardize the dataset. This step eliminates noise and enhances model performance.

1) Before Preprocessing: The dataset contains raw text with various inconsistencies, such as unnecessary punctuation, stopwords, and varying text formats. If left unprocessed, these inconsistencies can lead to poor model performance.

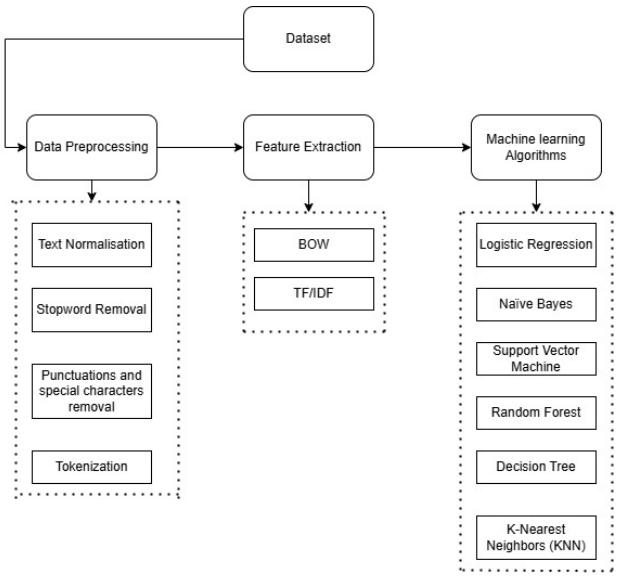

Fig. 1. Proposed System for Accuracy Prediction.

2) After Preprocessing: To improve accuracy, we apply the following preprocessing steps:

- **Text Normalization:** Convert all text to lowercase.
- **Stopword Removal:** Eliminate common words that do not add value to classification.
- **Punctuation and Special Character Removal:** Remove unnecessary symbols.
- **Tokenization:** Split text into individual words.

D. Feature Extraction

Since machine learning algorithms require numerical input, the textual content of tweets is converted into numerical feature vectors using two commonly adopted feature extraction techniques.

- **Bag of Words (BoW)** approach represents text by counting the frequency of individual words appearing in a tweet, without considering word order or semantic relationships. Although simple, this method is effective in capturing basic textual patterns.
- **Term Frequency–Inverse Document Frequency (TF-IDF)** technique assigns weights to words based on their importance within the dataset. It increases the weight of terms that are frequent in a specific tweet while reducing the impact of words that occur frequently across the entire corpus. This helps in emphasizing more informative terms and minimizing noise caused by commonly used words.

Both feature extraction methods are applied independently to analyze their effect on classification performance.

E. Model Training and Evaluation

We apply six classification algorithms to train and test our dataset:

- Support Vector Machine (SVM)
- Random Forest

- Logistic Regression
- Decision Tree
- Naïve Bayes
- K-Nearest Neighbors (KNN)

Each model is evaluated based on the following performance metrics:

(1) Accuracy: Accuracy measures the overall correctness of the model and is defined as the ratio of correctly predicted instances to the total number of predictions.

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \tag{1}$$

(2) Precision: Precision represents the proportion of correctly classified positive instances among all instances predicted as positive

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

(3) Recall: Recall indicates the proportion of correctly identified positive instances out of all actual positive instances.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

(4) F1 Score: F1-Score provides a balanced measure of model performance by computing the harmonic mean of precision and recall.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

1) Support Vector Machine (SVM): Achieved the highest accuracy of 94.65% for TF-IDF and 94.12% for BoW.

BoW:

TABLE III
EVALUATION RESULTS FOR SVM (BOW)

Accuracy	Precision	F1 Score	Recall
0.9412	0.94	0.94	0.95

Our BoW model gives an accuracy of 94.12%.

TF-IDF:

TABLE IV
EVALUATION RESULTS FOR SVM (TF-IDF)

Accuracy	Precision	F1 Score	Recall
0.9465	0.95	0.95	0.95

Our TF-IDF model gives an accuracy of 94.65%.

2) Random Forest: Performed slightly lower than SVM, with 94.59% accuracy.

BoW:

TABLE V
EVALUATION RESULTS FOR RANDOM FOREST (BOW)

Accuracy	Precision	F1-score	Recall
0.9452	0.95	0.94	0.94

Our BOW model yields an accuracy of 94.52%.

TF-IDF:

TABLE VI
EVALUATION RESULTS FOR RANDOM FOREST (TF-IDF)

Accuracy	Precision	F1-score	Recall
0.9459	0.94	0.95	0.94

Our TF-IDF model yields an accuracy of 94.59%.

3) Logistic Regression Model: Achieved 94.82% accuracy, proving effective for text classification.

BoW:

TABLE VII
EVALUATION RESULTS FOR LOGISTIC REGRESSION (BOW)

Accuracy	Precision	F1-score	Recall
0.9482	0.95	0.95	0.95

Our BoW model gives an accuracy of 94.82%.

TF-IDF:

TABLE VIII
EVALUATION RESULTS FOR LOGISTIC REGRESSION (TF-IDF)

Accuracy	Precision	F1-score	Recall
0.9331	0.93	0.93	0.93

Our TF-IDF model gives an accuracy of 93.31%.

4) Naïve Bayes Model: Performed well with 91.50% accuracy, but was outperformed by other models.

BoW:

TABLE IX
EVALUATION RESULTS FOR NAÏVE BAYES (BOW)

Accuracy	Precision	F1-score	Recall
0.9150	0.91	0.91	0.91

Our BoW model yields an accuracy of 91.50%.

TF-IDF:

TABLE X
EVALUATION RESULTS FOR NAÏVE BAYES (TF-IDF)

Accuracy	Precision	F1-score	Recall
0.8987	0.90	0.90	0.90

Our TF-IDF model yields an accuracy of 89.87%.

5) K-Nearest Neighbors (KNN) Model: Had the lowest accuracy of 84.17%, struggling with high-dimensional text data.

BoW:

TABLE XI
EVALUATION RESULTS FOR KNN (BOW)

Accuracy	Precision	F1-score	Recall
0.8417	0.86	0.84	0.85

Our BoW model yields an accuracy of 84.17%.

TF-IDF:

TABLE XII
EVALUATION RESULTS FOR KNN (TF-IDF)

Accuracy	Precision	F1-score	Recall
0.6327	0.77	0.60	0.66

Our TF-IDF model yields an accuracy of 63.27%.

6) Decision Tree Model: Performed best with BOW with an accuracy of 92.89

BoW:

TABLE XIII
EVALUATION RESULTS FOR DECISION TREE (BOW)

Accuracy	Precision	F1-score	Recall
0.9289	0.95	0.93	0.93

Our BoW model gives an accuracy of 92.89%.

TF-IDF:

TABLE XIV
EVALUATION RESULTS FOR DECISION TREE (TF-IDF)

Accuracy	Precision	F1-score	Recall
0.9278	0.93	0.93	0.93

Our TF-IDF model yields an accuracy of 92.78%.

F. Comparative Analysis

From our results, SVM emerged as the most accurate model, followed closely by Random Forest and Logistic Regression. KNN performed the worst due to its inefficiency in handling large textual data.

G. Future Scope

Future work will focus on improving classification performance by incorporating deep learning techniques such as Long Short-Term Memory (LSTM) networks and Transformer-based models, including BERT. These approaches can better capture contextual information in disaster-related text and enhance model robustness.

IV. RESULTS AND COMPARISON

In this study, disaster-related tweets were classified into on-topic and off-topic categories using two text vectorization techniques, namely Bag of Words (BoW) and Term Frequency–Inverse Document Frequency (TF-IDF). Six machine learning classifiers were evaluated, including Support Vector Machine (SVM), Random Forest, Logistic Regression, Naïve Bayes, K-Nearest Neighbors (KNN), and Decision Tree.

The experimental results show that SVM, Logistic Regression, and Random Forest perform better than the other classifiers for both BoW and TF-IDF features. The results obtained using BoW and TF-IDF are presented in Fig. 2 and Fig. 3, respectively, highlighting consistent performance trends across feature extraction methods.

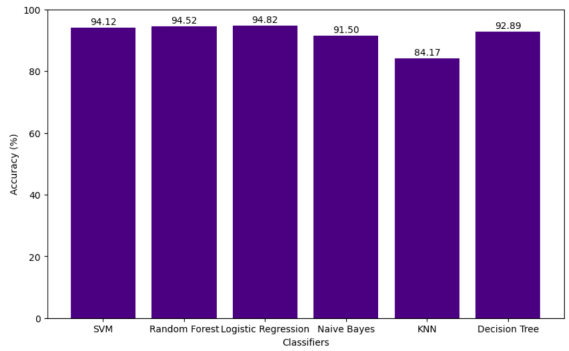

Fig. 2. Performance comparison of classifiers using BoW

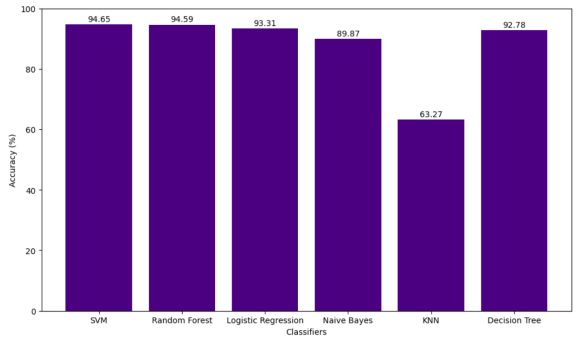

Fig. 3. Performance comparison of classifiers using TF-IDF

V. Conclusion

This research presented a machine learning-based approach for classifying disaster-related tweets into on-topic and off-topic categories using the CrisisLexT6 dataset. By applying Bag of Words (BoW) and Term Frequency–Inverse Document Frequency (TF-IDF) feature extraction techniques and evaluating six different classifiers, the study demonstrated that Support Vector Machine (94.65

The findings emphasize the role of machine learning-driven NLP techniques in extracting meaningful and actionable information from large volumes of social media data. Such automated systems can assist emergency responders and relief organizations by enabling faster identification of relevant information during disaster situations.

While traditional machine learning approaches have shown strong performance, future research will focus on incorporating deep learning models such as LSTMs and Transformer-based architectures, including BERT, to further improve accuracy and scalability. Additionally, extending the system to support real-time tweet classification and multilingual data could significantly enhance its applicability in global disaster management scenarios.

Overall, this study reinforces the importance of social media analytics in disaster management and highlights its potential to contribute toward more efficient, timely, and informed emergency response systems.

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